

BUA 304: Predictive Analytics: (Spring 2026)

Instructor: Rowaida Ali
E-mail: rowaida-ali@eru.edu.eg

Teaching Assistant: Mariam Abdelhamid
E-mail: mariam-abdelhamid@eru.edu.eg

Course Description: This course introduces foundations of predictive analytics and applied predictive modeling. Topics include basic predictive modeling methods for both classification and regression tasks, Time Series and Forecasting, predictive data mining, predictive spreadsheet models, and Monte Carlo simulation. Upon completion, students should be able to build and validate predictive models.

Pre-requisite(s): BA 201

Textbook:

Predictive Analytics: Parametric Models for Regression and Classification Using R. Tamhane A. John Wiley & Sons, Inc (2021).

Applied Predictive Analytics: Principles and Techniques for the Professional Data Analyst. John Wiley & Sons, Inc.

References:

- Business Analytics: Data Analysis and Decision Making, 7^{ed} Edition. S. Christian S and Wayne L. Cengage. (2020).
- Business Analytics Methods, Models, and Decisions, 2^{ed} Edition. James R. Pearson Education. (2017).
- Business Analytics Principles, Concepts, and Applications. Marc J, Dara G, and Christopher M. Pearson Education. (2014).
- Business Analytics: The Science of Data-Driven Decision Making. Kumar D. Wiley India Pvt. Ltd. (2017).
- Data Science for Business and Decision Making. Fa'vero L and Belfiore P. Elsevier Inc. (2019).
- Data Science for Business Professionals: A Practical Guide for Beginners, BPB Publications, India. Probyto Data Science and Consulting Pvt. Ltd (2020).
- Business Analytics, 3rd Edition. Jeffrey D, James J, Michael J, Jeffrey W, David R, Dennis J, Thomas A. Cengage Learning, Inc.(2019).

Learning Objectives:

By completing this course, students should be able to:

- ✚ Understand the theory and practice of different predictive modeling techniques, such as how to predict future outcomes based on historical data using regression, classification, and Time Series.

Grading:

Assignments	Lap	Quizzes	Project	Midterm	Final	Total
5%	5%	10%	10%	20%	50%	100%

Syllabus

- ✚ Foundations of Predictive Modeling with Regression
- ✚ Classification
- ✚ Time Series Analysis and Forecasting

Lec 1 Introduction to Predictive Analytics

Objectives

At the end of this lecture you should be able to:

- Understand predictive modeling problems requires knowledge of the business objectives.
- Know some specialized topics that are unique to multiple regression, include extra sums of squares, and the standardized version.
- detect violations of standard multiple linear regression assumptions.
- describe methods to address these violations through data transformations and other methods, also test the model assumptions.
- know other data problems that adversely affect regression results, namely, outliers, influential observations and multicollinearity.
- understand both graphical and formal statistical methods based on residuals.

Predictive Analytics

Predictive analytics consists of techniques that use past data to predict future events or ascertain the impact of one variable on another. These techniques include:

- Linear regression,
- time series analysis,
- some data-mining techniques,
- simulation.

Predictive Analytics Processing Steps: CRISP-DM

Stage	Description
Business Understanding	Define the project.
Data Understanding	Examine the data; identify problems in the data.
Data Preparation	Fix problems in the data; create derived variables.
Modeling	Build predictive or descriptive models.
Evaluation	Assess models; report on the expected effects of models.
Deployment	Plan for use of models.

Case Study: Invoice Fraud Detection

The organization in question was responsible for reviewing invoices for service payments, some of which were fraudulently submitted. Due to resource limitations, it was not feasible to investigate all invoices. Thus, a predictive model was proposed to identify suspicious invoices for further investigation.

Business Objectives

Defining "fraud" was a critical step in the modeling process. Two main definitions were considered:

- **Strict Definition:** An invoice is considered fraudulent only if it has been prosecuted and the payee convicted of fraud.
- **Broader Definition:** An invoice is considered fraudulent if it is flagged as suspicious by auditors, even if no definitive proof exists.

Comparison of Definitions

- **Strict Definition:** Ensures high accuracy but results in fewer labeled fraudulent cases.
- **Broader Definition:** Increases the number of flagged cases but may lead to false positives (non-fraudulent invoices being misclassified).

The strict definition was chosen for this project to ensure data and model accuracy.

Project Data

- **Dataset Size:** Hundreds of thousands of invoices, but only a limited number were labeled as fraudulent due to the strict definition.
- **Classification Challenge:** Some fraudulent invoices might remain undetected, leading to imbalanced data that could impact the model's performance.

Fraud Assumption: The fraud rate was assumed to be 1%, but only half of the fraudulent invoices were identified and prosecuted.

Addressing Data Imbalance

A balanced sampling approach was adopted to reduce the likelihood of including mislabeled invoices, improving the model's predictive accuracy.

Target Variables

The main target variable was a binary classification: Fraudulent (1) or Non-Fraudulent (0).

After cluster analysis, four distinct fraud types were identified.

The final classification system included five categories: four fraud types and one non-fraud type.

Modeling Objectives

- Develop multi-class classification models to detect four fraud types.
- Minimize misclassification errors, especially false positives.
- Improve audit efficiency by identifying the top 100 high-risk invoices each month.

Model Selection and Evaluation Criteria

Various models were tested, focusing on achieving a balance between accuracy and operational efficiency.

Key Selection Criteria

- Reducing false positives: Misclassifying a non-fraudulent invoice as fraudulent was penalized more than the opposite.
- Maximizing investigator efficiency: The model was evaluated based on how many actual fraud cases were included in the top 100 flagged invoices each month.

Model Deployment

- The model was run monthly on all invoices.
- The top 100 highest-scoring invoices were flagged for auditors.
- Results were analyzed and model updates were made periodically to adapt to changing fraud patterns.

Modeling

Suppose that we observe a quantitative response Y and p different predictors, $X_1, X_2, \dots, X_p$. We assume that there is some relationship between Y and $X = (X_1, X_2, \dots, X_p)$, which can be written in the very general form:

$$Y = f(X) + \epsilon$$

Here f is some fixed but unknown function of $X_1, X_2, \dots, X_p$, and ϵ is a random error term, which is independent of X and has mean zero. In this formulation, f represents the systematic information that X provides about Y .

Why Estimate f ?

There are two main reasons that we may wish to estimate f : *prediction* and *inference*.

Prediction

In pure prediction problems, we usually treat $\hat{f}$ as a black box. We don't care how it works we only care that it gives *accurate predictions*.

How accurate can our prediction $\hat{Y}$ be?

The expected prediction error decomposes into two parts:

$$\begin{aligned} E(Y - \hat{Y})^2 &= E[f(X) + \epsilon - \hat{f}(X)]^2 \\ &= \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{Reducible}} + \underbrace{\text{Var}(\epsilon)}_{\text{Irreducible}} \end{aligned}$$

Reducible error: We can reduce this by getting a better estimate $\hat{f}$ better model, more data, better features, tuning, etc.

Irreducible error: This part cannot be reduced, no matter how good $\hat{f}$ is.

Inference

Understand how Y is related to $X_1, X_2, \dots, X_p$. We still estimate f , but now $\hat{f}$ cannot be a black box. We need to know and interpret its exact form.

Common inference questions:

- Which predictors are actually associated with Y ? (Variable importance / selection)
- What is the direction and strength of the relationship? (Positive? Negative? How much does Y change when X increases?)
- Does the relationship depend on other variables? (Interactions)
- Is the relationship linear, or is it more complicated?

“Inference is about understanding, not just predicting.

- Here interpretability is more important than raw accuracy.”

Many real problems are a mix.

How Do We Estimate f ?

- we want to find a function $\hat{f}$ such that $Y \approx \hat{f}(X)$ for any observation (X, Y) .
- most methods for this task can be characterized as either *parametric* or *non-parametric*.

Supervised versus Unsupervised Learning

Supervised Learning

- In supervised learning, one of the variables is designated as a *response*, *outcome*, *dependent*, or *output* variable.
- The goal is to model its relationship with the so-called predictor, covariate, independent or input variables (called features in machine learning).
- Some examples of supervised learning are the following:
 - predicting customer purchase behavior as a function of past purchase history and demographic and socioeconomic data;
 - predicting the selling price of a house as a function of floor area, plot size, amenities, school district and economic indicators;
 - classifying a patient's disease status based on a battery of laboratory tests
 - classifying a bank loan as in good standing or likely to default as a function of borrower's financial parameters such as income, debt etc.

Unsupervised Learning

- In unsupervised learning, there is no designated response variable, all variables being of the same genre.
- The goal is to uncover relationships, associations, and patterns that underlie the data.
- Some examples are the following:
 - determination of latent factors in multivariate data and market segmentation, e.g. clustering of customers with similar attributes.

Parametric Versus Nonparametric Models

	Parametric Models	Nonparametric Models
The functional form	specified; only the parameters of the model are unknown and are estimated from data.	either not specified or is too complex to be expressed in a closed form involving a few parameters
Examples of supervised learning	- multiple regression, - logistic regression, - discriminant analysis, - generalized linear models, - the Cox proportional hazards model for survival data	- classification and regression trees (CART), - random forests, - neural nets (NN), - support vector machines (SVM). Technically, NN and SVM are complex parametric models
Examples of unsupervised learning	- correlation analysis, - principal components analysis, - factor analysis.	- cluster analysis - association rules.

The Trade-Off Between Prediction Accuracy and Model Interpretability



A representation of the tradeoff between flexibility and interpretability, using different statistical learning methods. In general, as the flexibility of a method increases, its interpretability decreases.

Multiple linear regression: basics

Multiple regression is a basic and essential tool of predictive modeling and serves as a foundation for more advanced techniques.

Multiple linear regression model

Model in scalar notation

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i \quad (i = 1, \dots, n).$$

Where,

- $\beta_0, \beta_1, \dots, \beta_p$ are unknown parameters, β_0 is the intercept or the constant term, $\beta_1, \dots, \beta_p$ are the regression coefficients
- the ε_i are independent and identically distributed (i.i.d.) $N(0, \sigma^2)$ random errors.
- From this model, it follows that the y_i are independent $N(\mu_i, \sigma^2)$, where

$$\mu_i = E(y_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} \quad (i = 1, \dots, n).$$

Note the following points about this model:

- This is called a linear model because it is linear in the β parameters - not necessarily in the x 's.
- Any nonlinear functions of the x 's, x^2 , $\log x$, or $x_1 x_2$, may be used as predictors, and the model is still regarded as linear.
- The reason that linearity in the β 's is critical is that it makes the equations for finding their OLS estimators linear, which makes them easy to solve with closed form solutions.
- Furthermore, these OLS estimators are then linear functions of the responses y_i 's, which makes their sampling distributions simple and so inferences on them straightforward.
- Some nonlinear models can be transformed into linear models.
- nonlinear model is nonlinear in the β 's and cannot be transformed into a linear form. An example from chemical process kinetics is the model

$$y = \frac{\beta_1}{\beta_1 - \beta_2} [e^{-\beta_2 t} - e^{-\beta_1 t}] + \varepsilon.$$

where y is the percent reaction completed, t is the reaction time, and β_1 and β_2 are reaction rate constants.

No transformation can linearize this model and nonlinear regression techniques must be used to estimate the rate constants.

Model in matrix notation

$$y = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

Where,

- $\mathbf{y}$ is the response vector, $\mathbf{X}$ is called the model matrix, $\boldsymbol{\beta}$ is the parameter vector, and $\boldsymbol{\varepsilon}$ is the random error vector.
- ε_i are i.i.d. $N(0, \sigma^2)$ is that the random vector $\boldsymbol{\varepsilon}$ has an n -variate normal distribution with mean vector 0 (the null vector of all 0's) and covariance matrix $\sigma^2 \mathbf{I}$, where $\mathbf{I}$ is an $n \times n$ identity matrix. The diagonal entries of $\sigma^2 \mathbf{I}$ are the variances σ^2 of the y_i 's and the off-diagonal entries are the covariances between the pairs y_i and y_j , which are all 0 under the assumption of independence among the random errors ε_i 's.
- The distribution of the response vector $\mathbf{y}$ is then the n -variate normal distribution with mean vector $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ and covariance matrix $\sigma^2 \mathbf{I}$. $E(\mathbf{y}) = \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ where $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)'$

Fitted values and residuals

The fitted values are defined as

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \text{ where, } \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

The residuals are defined as

$$e_i = y_i - \hat{y}_i \quad (i = 1, \dots, n),$$

and the residual vector is defined as:

$$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \mathbf{y} - \hat{\mathbf{y}}.$$

- The ε_i are unobservable random differences between the y_i and $E(y_i) = \mu_i = \mathbf{x}_i' \boldsymbol{\beta}$ (unobservable because $E(y_i)$ is unknown),
- the e_i are observable (because they are computable) differences between the observed y_i and the fitted $\hat{y}_i$, where $\hat{y}_i = \hat{\mu}_i = \mathbf{x}_i' \hat{\boldsymbol{\beta}}$.

In matrix notation, we can write the random error vector and the residual vector as

$$\boldsymbol{\varepsilon} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta} \quad \text{and} \quad \mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Sequential (or Extra) Sums of Squares

An extra sum of squares measures the marginal reduction in the error sum of squares when one or several predictor variables are added to the regression model, given that other predictor variables are already in the model. It can be viewed in either of two ways:

- It is the reduction in the error sum of squares (SSE) when one or more predictor variables are added to the model.
- Or, it is the increase in the regression sum of squares (SSR) when one or more predictor variables are added to the model.

Let: $\text{SSE}(\mathbf{X}_1)$ and $\text{SSR}(\mathbf{X}_1)$ be the error and regression sums of squares obtained from fitting the simple linear regression model. Let

$$\text{SSE}(\mathbf{X}_1, \mathbf{X}_2) \text{ and } \text{SSR}(\mathbf{X}_1, \mathbf{X}_2)$$

be the error and regression sums of squares obtained from fitting the multiple regression model with both $\mathbf{X}_1$ and $\mathbf{X}_2$.

Define the extra sum of squares $\text{SSR}(\mathbf{X}_2 | \mathbf{X}_1)$ to be the reduction in the error sum of squares resulting from adding $\mathbf{X}_2$ to the model that already includes $\mathbf{X}_1$:

$$\text{SSR}(\mathbf{X}_2 | \mathbf{X}_1) = \text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X}_1, \mathbf{X}_2)$$

$SSR(X_2 | X_1)$ tells us how much of the Y variation not already explained by X_1 can be explained by X_2 . Intuitively, $SSR(X_2 | X_1)$ provides an indication of whether it's useful to add X_2 to a model that already includes X_1 .

This idea of an extra sum of squares can be extended to the case in which more than two predictors are already in the model. For example, the extra sum of squares

$$SSR(X_3 | X_1, X_2) = SSE(X_1, X_2) - SSE(X_1, X_2, X_3)$$

tells us how much of the Y variation not already explained by X_1 and X_2 can be explained by X_3 . It provides an indication of whether it's useful to add X_3 to a model that already includes X_1 and X_2 .

Uses of Extra Sums of Squares in Tests for Regression Coefficients

- The extra sum of squares method is used when we want to test hypotheses on multiple β 's simultaneously.
- One example of this is when we want to test whether the full model with all predictors fits the data significantly better than a partial or reduced model consisting of a subset of predictors. In other words, whether adding the extra predictors improves the fit significantly.
- The partial model is said to be nested under the full model.

Coefficients of Partial Determination

Extra sums of squares are not only useful for tests on the regression coefficients of a multiple regression model, but they are also encountered in descriptive measures of relationship called coefficients of partial determination.

Let $SSE(x_1)$ and $SSE(x_1, x_2)$ denote the SSEs from regressions of y on x_1 and on x_1, x_2 , respectively. Then

$$r_{y x_2 | x_1}^2 = \frac{SSE(x_1) - SSE(x_1, x_2)}{SSE(x_1)} = 1 - \frac{SSE(x_1, x_2)}{SSE(x_1)}.$$

and $r_{y x_2 | x_1}$ is the square root of the above, its sign being that of $\hat{\beta}_2$ in the regression of y on x_1, x_2 . Thus, $r_{y x_2 | x_1}^2$ is the additional relative contribution to reduction in SSE due to x_2 compared to the SSE with only x_1 in the model. This formula extends the definition $R^2 = 1 - SSE/SST$ by replacing SSE with $SSE(x_1, x_2)$ and SST with $SSE(x_1)$. Essentially, R^2 compares the SSE of the full

model consisting of p variables with $SSE = SST$ of the null model consisting only of the intercept term, which is treated as the partial model. Here $\{x_1, x_2\}$ is the full model and $\{x_1\}$ is the partial model.

Partial correlation coefficients

A different method computing $r_{yx_2|x_1}$ is from the bivariate correlations r_{yx_1} , r_{yx_2} , and $r_{x_1x_2}$ using the formula:

$$r_{yx_2|x_1} = \frac{r_{yx_2} - r_{yx_1}r_{x_1x_2}}{\sqrt{(1-r_{yx_1}^2)(1-r_{x_1x_2}^2)}}.$$

Test of significance of partial correlation coefficient

$$F_p = \frac{r_{yx_p|x_1, \dots, x_{p-1}}^2 [n-(p+1)]}{1-r_{yx_p|x_1, \dots, x_{p-1}}^2}.$$

The corresponding t -statistic is the square root of F_p :

$$t_p = \sqrt{F_p} = \frac{r_{yx_p|x_1, \dots, x_{p-1}} \sqrt{n-(p+1)}}{\sqrt{1-r_{yx_p|x_1, \dots, x_{p-1}}^2}}.$$

We conclude that $r_{yx_p|x_1, \dots, x_{p-1}}$ is significant at level α if $F_p > f_{1, n-(p+1), \alpha}$ or equivalently if $|t_p| > t_{n-(p+1), \alpha/2}$.

we see that it is a generalization of the test on the bivariate correlation coefficient.

Example 1: College admission committees are faced with the task of who to admit from thousands of applicants. There are many considerations including their college entrance test scores, high school grade point average (GPA) and rank, essays, extracurricular activities, recommendation letters, and so on. In this example, we will look at predicting just one measure of academic success in college, namely graduating GPA. We will consider two predictors of that success, namely, college entrance Verbal and Math test scores. Data on 40 students are shown in Table 2.1. Some questions of interest are the following:

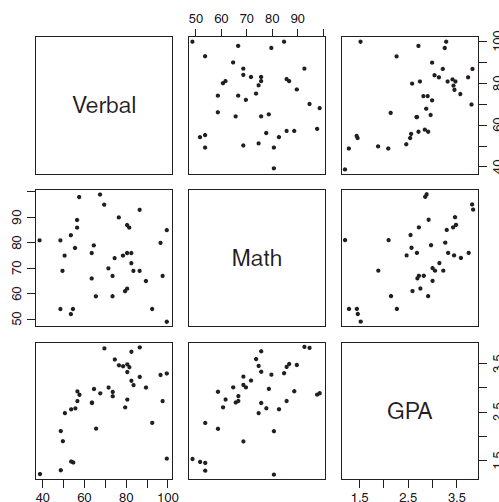
1. what is the relative importance of each test score in predicting the GPA
2. what is a good prediction model for GPA in terms of these two predictors.

Table 2.1. Entrance test scores and graduating GPA

Verbal	Math	GPA	Verbal	Math	GPA	Verbal	Math	GPA
81	87	3.49	83	76	3.75	97	80	3.27
68	99	2.89	64	66	2.70	77	90	3.47
57	86	2.73	83	72	3.15	49	54	1.3
100	49	1.54	93	54	2.28	39	81	1.22
54	83	2.56	74	59	2.92	87	69	3.23
82	86	3.43	51	75	2.48	70	95	3.82
75	74	3.59	79	75	3.45	57	89	2.93
58	98	2.86	81	62	2.76	74	67	2.83
55	54	1.46	50	69	1.90	87	93	3.84
49	81	2.11	72	70	3.01	90	65	3.01
64	76	2.69	54	52	1.48	81	76	3.33
66	59	2.16	65	79	2.98	84	69	3.06
80	61	2.60	56	78	2.58			
100	85	3.30	98	67	2.73			

Before running a regression of GPA on Verbal and Math scores, it is useful to study relationships among them by making scatter plots and computing the correlation matrix.

Matrix scatter plot between GPA, Verbal score and Math score.



We can see from this plot that Verbal and Math scores are fairly uncorrelated with each other while both are moderately correlated with GPA. Thus, the two predictors make relatively independent and roughly equal contributions to the GPA and it makes sense to fit a linear model for GPA that includes both of them. The above visual impressions based on the matrix scatter plot are confirmed by the correlation matrix below.

	Verbal	Math	GPA
Verbal	1	-0.107	0.529
Math	-0.107	1	0.573
GPA	0.529	0.573	1

The regression equation fitted

$$\widehat{GPA} = -1.5705 + 0.0257Verbal + 0.0336Math.$$

ANOVA table for linear regression of GPA on entrance test scores

Source	SS	d.f.	MS	F	P
Regression	12.7859	2	6.3930	39.51	0.000
Error	5.9876	37	0.1618		
Total	18.7735	39			

A question that is often of interest in such regression problems is what proportion of this SS can be attributed to each predictor, in this case to Verbal and Math scores????

Using the extra SS method, we can compute the SS due to each predictor by taking the difference between SSR for the full model and SSR0 for the partial model, which omits the predictor of interest.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Verbal	1	5.2549	5.2549	32.472	1.608e-06 ***
Math	1	7.5311	7.5311	46.538	4.902e-08 ***
Residuals	37	5.9876	0.1618		

In this table, the SSs for Verbal and Math do add up to SSR as can be checked from

$$SS_{\text{Verbal}} + SS_{\text{Math}} = 5.2549 + 7.5311 = 12.7860$$

These SSs are referred to as **sequential sums of squares** or **type I sums of squares**, and it is easy to see that they always add up to SSR. But they depend on the order in which the predictors are entered into the regression model; if the order is changed, then they change as well.

Consider comparing a first-degree model versus a second-degree model in two variables to see if it provides a significantly better fit.

The fitted quadratic model is

$$\widehat{\text{GPA}} = -11.458 + .189\text{Verbal} + 0.159\text{Math} - 0.0011\text{Verbal}^2 - 0.0009\text{Math}^2.$$

To check if the two quadratic terms significantly improve the fit, we can do the extra SS test as follows:

1. Taking the linear model as the partial model

$$\widehat{\text{GPA}} = -1.5705 + 0.0257\text{Verbal} + 0.0336\text{Math}$$

we get $\text{SSE}_0 = 5.9876$

2. Taking the quadratic model as the full model,

ANOVA table for GPA versus entrance test scores: quadratic regression

Source	SS	d.f.	MS	F	P
Regression	17.4845	4	4.371	118.69	0.000
Residual error	1.2890	35	0.037		
Total	18.7735	39			

$$\text{SSE} = 1.2890 .$$

3. the F -statistic for testing $\beta_3 = \beta_4 = 0$ in the quadratic model equals

$$F = \frac{(5.9876 - 1.2890)/2}{0.0368} = 63.79,$$

with 2 and 35 d.f., which is clearly highly significant ($P < 0.001$).

Partial Correlation Coefficients:

GPA = y , Verbal = x_1 , and Math = x_2 , we want to calculate $r_{yx_1|x_2}$ and $r_{yx_2|x_1}$.

By running separate regressions of GPA on Verbal and on Math, we get $SSE(x_1) = 13.5186$ and $SSE(x_2) = 12.6063$. $SSE(x_1, x_2) = 5.9876$. Hence,

$$r_{yx_1|x_2}^2 = 1 - \frac{5.9876}{12.6063} = 0.5250 \text{ and } r_{yx_2|x_1}^2 = 1 - \frac{5.9876}{13.5186} = 0.5571.$$

Thus, $r_{yx_1|x_2} = \sqrt{0.5250} = 0.7246$ and $r_{yx_2|x_1} = \sqrt{0.5571} = 0.7464$.

The coefficients of both are positive since the corresponding regression coefficients are positive.

The same values can be obtained using the formula $r_{yx_2|x_1} = \frac{r_{yx_2} - r_{yx_1}r_{x_1x_2}}{\sqrt{(1-r_{yx_1}^2)(1-r_{x_1x_2}^2)}}$

The bivariate correlations are $r_{yx_1} = 0.5291$, $r_{yx_2} = 0.5732$, and $r_{x_1x_2} = -0.1069$. By substituting these values, we get

$$r_{yx_1|x_2} = \frac{0.5291 + 0.1069 \times 0.5732}{\sqrt{(1-0.1069^2)(1-0.5732^2)}} = 0.7246.$$

and

$$r_{yx_2|x_1} = \frac{0.5732 + 0.1069 \times 0.5291}{\sqrt{(1-0.1069^2)(1-0.5291^2)}} = 0.7464.$$

The t -statistics to test the significance of $r_{yx_1|x_2}$ and $r_{yx_2|x_1}$ are

$$t_{yx_1|x_2} = \frac{0.7246\sqrt{37}}{\sqrt{1-0.7246^2}} = 6.395 \text{ and } t_{yx_2|x_1} = \frac{0.7464\sqrt{37}}{\sqrt{1-0.7464^2}} = 6.822$$

College GPA and Entrance Test Scores: Interaction Effect

we fitted a quadratic model to the GPA data which we found to give a significantly better fit than the linear model. In this example, we will add the interaction term to see whether it further improves the fit of the model significantly. Thus consider the full second-degree model:

ANOVA table for GPA versus entrance test scores: quadratic regression with interaction

Source	SS	d.f.	MS	F	P
Regression	17.5827	5	3.5165	100.41	0.000

Residual error	1.1908	34	0.0350
Total	18.7735	39	

Then the F -statistic for $H_0: \beta_5 = 0$ equals

$$F = \frac{(1.2890 - 1.1908)/1}{1.1908/34} = 2.806,$$

which for $\alpha = 0.10$ can be compared with $f_{1,34,0.10} = 2.859$. Since $F = 2.806 < 2.859$, we fail to reject H_0 .

The equivalence between the t -test on Verbal $\times$ Math interaction in the R output and the F -test calculated using the extra SS method can be verified by checking that $t = \sqrt{2.806} = 1.675$ and $t_{34,0.05} = \sqrt{f_{1,34,0.10}} = \sqrt{2.859} = 1.691$.

Coefficients:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.9167631  1.3544134  -7.322 1.75e-08 ***
Verbal       0.1668098  0.0212447   7.852 3.85e-09 ***
Math        0.1375972  0.0267340   5.147 1.11e-05 ***
I(Verbal ^ 2) -0.0011082  0.0001173  -9.449 4.88e-11 ***
I(Math ^ 2)   -0.0008433  0.0001594  -5.290 7.23e-06 ***
Verbal:Math   0.0002411  0.0001440   1.675  0.103
---
Residual standard error: 0.1871 on 34 degrees of freedom
Multiple R-squared:  0.9366, Adjusted R-squared:  0.9272
F-statistic: 100.4 on 5 and 34 DF,  p-value: < 2.2e-16

```

Standardized Multiple Regression Model

A standardized form of the general multiple regression model is employed to control roundoff errors in normal equations calculations and to permit comparisons of the estimated regression coefficients in common units.

standardized regression coefficients are unitless and hence can be compared with each other. They are also useful for computational purposes as we will explain below.

$$b_k = \left(\frac{S_Y}{S_k} \right) b_k^* \quad (k = 1, \dots, p-1)$$
$$b_0 = \bar{Y} - b_1 \bar{X}_1 - \dots - b_{p-1} \bar{X}_{p-1}$$

Model Assumptions and Distribution of Residuals

The following assumptions underlie the multiple regression model, in particular about the random errors ε_i .

1. **Normality:** The ε_i are normally distributed.
2. **Homoscedasticity:** The ε_i have a constant variance σ^2 .
3. **Independence:** The ε_i are statistically independent.
4. **No outliers:** No observations deviate significantly from the specified model.
5. **The model is correctly specified.**

In many cases, visually assessing residual plots, histograms, and QQ-plots to check the four multiple linear regression assumptions is sufficient. However, in ambiguous cases or when we wish to complement our visual conclusions with quantitative evidence, there are a number of tests available that address each of the assumptions.

Lack of Fit Test (LOF)

- LOF test is a formal test for determining whether a specific type of regression function adequately fits the data
- We illustrate this test for ascertaining whether a linear regression function is a good fit for the data.

Assumptions

The lack of fit test assumes that the observations Y for given X are:

- Independent
- normally distributed,
- the distributions of Y have the same variance σ^2 .
- The lack of fit test requires repeat, observations at one or more X levels.

Example 2: In an experiment involving 12 similar but scattered suburban branch offices of a commercial bank, holders of checking accounts at the offices were offered gifts for setting up money market accounts. Minimum initial deposits in the new money market account were specified to qualify for the gift. The value of the gift was directly proportional to the specified minimum deposit. Various levels of

minimum deposit and related gift values were used in the experiment in order to ascertain the relation between the specified minimum deposit and gift value, on the one hand, and number of accounts opened at the office, on the other. Altogether, six levels of minimum deposit and proportional gift value were used, with two of the branch offices assigned at random to each level. One branch office had a fire during the period and was dropped from the study. The following table contains the results, where X is the amount of minimum deposit and Y is the number of new money market accounts that were opened and qualified for the gift during the test period.

Branch i	Size of Minimum Deposit (dollars) X_i	Number of New Accounts Y_i	Branch i	Size of Minimum Deposit (dollars) X_i	Number of New Accounts Y_i
1	125	160	7	75	42
2	100	112	8	175	124
3	200	124	9	125	150
4	75	28	10	200	104
5	150	152	11	100	136
6	175	156			

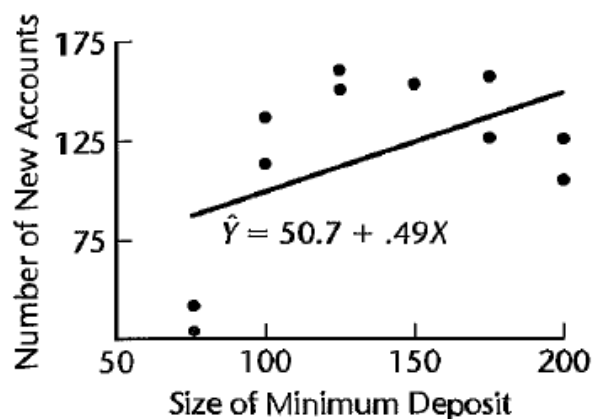
A linear regression function was fitted in the usual fashion; it is:

$$\hat{Y} = 50.72251 + 0.48670X$$

ANOVA Table

Source of Variation	SS	df	MS
Regression	5,141.3	1	5,141.3
Error	14,741.6	9	1,638.0
Total	19,882.9	10	

Scatter Plot and Fitted Regression Line



Data Arranged by Replicate Number

Replicate	Size of Minimum Deposit (dollars)					
	$j = 1$ $x_1 = 75$	$j = 2$ $x_2 = 100$	$j = 3$ $x_3 = 125$	$j = 4$ $x_4 = 150$	$j = 5$ $x_5 = 175$	$j = 6$ $x_6 = 200$
$j = 1$	28	112	160		156	124
$j = 2$	42	136	150	152	124	104
Mean $\bar{Y}_j$	35	124	155	152	140	114

The general linear test requires consideration of the reduced model under H_0 . For testing the appropriateness of a linear regression relation, the alternatives are:

$$H_0: E\{Y\} = \beta_0 + \beta_1 X$$

$$H_1: E\{Y\} \neq \beta_0 + \beta_1 X$$

The reduced model under H_0 therefore is:

$$Y_{ij} = \beta_0 + \beta_1 X_j + \varepsilon_{ij} \text{ Reduced model}$$

General ANOVA Table for Testing Lack of Fit of Simple Linear Regression Function and ANOVA

Source of Variation	SS	df	MS	F
Regression	$SSR = \sum \sum (\hat{Y}_{ij} - \bar{Y})^2$	1	$MSR = \frac{SSR}{1}$	MSR/MSE
Error	$SSE = \sum \sum (Y_{ij} - \hat{Y}_{ij})^2$	$n - 2$	$MSE = \frac{SSE}{n - 2}$	
Lack of fit	$SSLF = \sum \sum (\bar{y}_j - \hat{y}_{ij})^2$	$c - 2$	$MSLF = \frac{SSLF}{c - 2}$	$MSLF/MSPE$
Pure error	$SSPE = \sum \sum (Y_{ij} - \bar{y}_j)^2$	$n - c$	$MSPE = \frac{SSPE}{n - c}$	
Total	$SSTO = \sum \sum (Y_{ij} - \bar{Y})^2$	$n - 1$		

Decomposing the Error when There are Replicates:

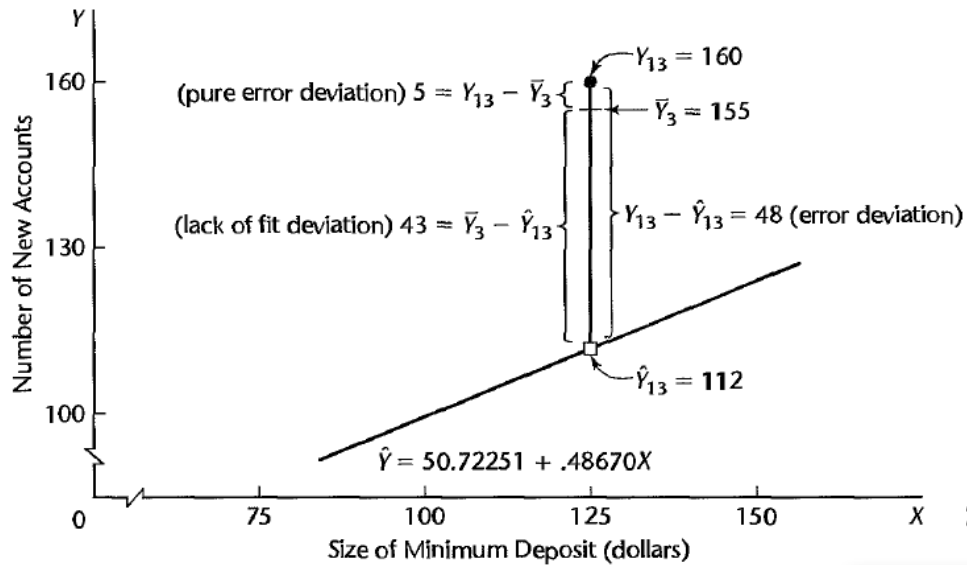
- We break down the residual error ("error sum of squares" — denoted SSE) into two components:
 - a component that is due to lack of model fit ("lack of fit sum of squares" — denoted SSLF)
 - a component that is due to pure random error ("pure error sum of squares" — denoted SSPE)
- If the lack of fit sum of squares is a large component of the residual error, it suggests that a linear function is inadequate.

$$SSE = SSPE + SSLF$$

This decomposition follows from the identity:

$$\underbrace{Y_{ij} - \hat{Y}_{ij}}_{\text{Error deviation}} = \underbrace{Y_{ij} - \bar{Y}_j}_{\text{Pure error deviation}} + \underbrace{\bar{Y}_j - \hat{Y}_{ij}}_{\text{Lack of fit deviation}}$$

Illustration of Decomposition of Error Deviation



Source of Variation	SS	df	MS	F	p
Regression	5,141.3	1	5141	3.14	0.11
Error	14,741.6	9	1638		
Lack of fit	13,593.6	4	3398	14.8	0.006
Pure error	1,148.0	5	230		
Total	19,882.9	10			

You might notice that the lack of fit F -statistic is calculated by dividing the lack of fit mean square ($MSLF = 3398$) by the pure error mean square ($MSPE = 230$) to get 14.80. How do we know that this F -statistic helps us in testing the hypotheses:

- H_0 : The relationship assumed in the model is reasonable, i.e., there is no lack of fit.
- H_1 : The relationship assumed in the model is not reasonable, i.e., there is lack of fit.

The answer lies in the "expected mean squares." In our sample of $n = 11$ newly opened checking accounts, we obtained $MSLF = 3398$. If we had taken a different

random sample of size $n = 11$, we would have obtained a different value for $MSLF$. Theory tells us that the average of all of the possible $MSLF$ values we could obtain is:

$$E(MSLF) = \sigma^2 + \frac{\sum n_i(\mu_i - (\beta_0 + \beta_1 X_i))^2}{c - 2}$$

That is, we should expect $MSLF$, on average, to equal the above quantity $-\sigma^2$ plus another messy-looking term. Think about that messy term. If the null hypothesis is true, i.e., if the relationship between the predictor x and the response y is linear, then μ_i equals $\beta_0 + \beta_1 X_i$ and the messy term becomes 0 and goes away. That is, if there is no lack of fit, we should expect the lack of fit mean square $MSLF$ to equal σ^2 .

What should we expect $MSPE$ to equal? Theory tells us it should, on average, always equal σ^2 :

$$E(MSPE) = \sigma^2$$

The logic behind the calculation of the F -statistic is now clear:

- If there is a linear relationship between x and y , then $\mu_i = \beta_0 + \beta_1 X_i$. That is, there is no lack of fit in the simple linear regression model. We would expect the ratio $MSLF/MSPE$ to be close to 1 .
- If there is not a linear relationship between x and y , then $\mu_i \neq \beta_0 + \beta_1 X_i$. That is, there is lack of fit in the simple linear regression model. We would expect the ratio $MSLF/MSPE$ to be large, i.e., a value greater than 1 .

Checking normality

- To test the normality assumption on the ε_i 's, the recommended method is the normal quantile–quantile plot (called the normal Q–Q plot or simply the normal plot) of the residuals.
- A histogram is not particularly useful for checking normality as its closeness to a normal bell-shaped curve is difficult to judge by eye, and there are other distributions that are also bell-shaped, e.g. the t-distribution and the Cauchy distribution.
- There are several formal normality tests

Tests for Normality

Goodness of fit tests can be used for examining the normality of the error terms. For instance, the chi-square test or the Kolmogorov-Smirnov test and its modification, the Lilliefors test, can be employed for testing the normality of the error terms by analyzing the residuals.

For each test discussed below, the formal hypothesis test is written as:

H_0 : the errors follow a normal distribution

H_1 : the errors do not follow a normal distribution.

Anderson-Darling Test

The Anderson-Darling Test measures the area between a fitted line (based on the chosen distribution) and a nonparametric step function (based on the plot points). The statistic is a squared distance that is weighted more heavily in the tails of the distribution. Smaller Anderson-Darling values indicate that the distribution fits the data better. The test statistic is given by:

$$A^2 = -n - \sum_{i=1}^n \frac{2i-1}{n} [\log F(e_i) + \log (1 - F(e_{n+1-i}))].$$

where $F(\cdot)$ is the cumulative distribution of the normal distribution. The test statistic is compared against the critical values from a normal distribution in order to determine the p-value.

Shapiro-Wilk Test

The Shapiro-Wilk Test uses the test statistic

$$W = \frac{(\sum_{i=1}^n a_i e_{(i)})^2}{\sum_{i=1}^n (e_i - \bar{e})^2}.$$

where the a_i values are calculated using the means, variances, and covariances of the $e_{(i)}$. W is compared against tabulated values of this statistic's distribution. Small values of W will lead to rejection of the null hypothesis.

Ryan-Joiner Test

The Ryan-Joiner Test is a simpler alternative to the Shapiro-Wilk test. The test statistic is actually a correlation coefficient calculated by

$$R_p = \frac{\sum_{i=1}^n e_{(i)} z_{(i)}}{\sqrt{s^2(n-1) \sum_{i=1}^n z_{(i)}^2}}.$$

where the $z_{(i)}$ values are the z-score values (i.e., normal values) of the corresponding $e_{(i)}$ value and s^2 is the sample variance. Values of R_p closer to 1 indicate that the errors are normally distributed.

Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov Test (also known as the Lilliefors Test) compares the empirical cumulative distribution function of sample data with the distribution expected if the data were normal. If this observed difference is sufficiently large, the test will reject the null hypothesis of population normality. The test statistic is given by:

$$D = \max(D^+, D^-),$$

Where, $D^+ = \max_i (i/n - F(e_{(i)}))$, $D^- = \max(F(e_{(i)}) - (i-1)/n)$.

Tests for Constant Error Variance

F-Test

Suppose we partition the residuals of observations into two groups - one consisting of residuals associated with the lowest predictor values and the other consisting of those belonging to the highest predictor values. Treating these two groups as if they could (potentially) represent two different populations, we can test

$$\begin{aligned} H_0: \sigma_1^2 &= \sigma_2^2 \\ H_1: \sigma_1^2 &\neq \sigma_2^2 \end{aligned}$$

using the F -statistic $F^* = s_1^2/s_2^2$. This test statistic is distributed according to a F_{n_1-1, n_2-1} distribution, so if $F^* \geq F_{n_1-1, n_2-1; 1-\alpha}$, then reject the null hypothesis and conclude that there is statistically significant evidence that the variance is not constant.

Modified Levene Test (Brown-Forsythe test)

- Another test for nonconstant variance is the modified Levene test (sometimes called the Brown-Forsythe test). This test does not require the error terms to be drawn from a normal distribution and hence it is a nonparametric test.

- The test is constructed by grouping the residuals into g groups according to the values of the quantity on the horizontal axis of the residual plot. It is typically recommended that each group has at least 25 observations and usually $g = 2$ groups are used.
- Begin by letting group 1 consist of the residuals associated with the n_1 lowest values of the predictor. Then, let group 2 consists of the residuals associated with the n_2 highest values of the predictor (so $n_1 + n_2 = n$).
- The objective is to perform the following hypothesis test:

H_0 : the variance is constant

H_A : the variance is not constant.

- The test statistic for the above is computed as follows:

$d_{i,j} = |e_{i,j} - \tilde{e}_i|$, where $\tilde{e}_i$ is the median of the i^{th} group of residuals.

$$s_L = \sqrt{\frac{\sum_{j=1}^{n_1} (d_{1,j} - \bar{d}_1)^2 + \sum_{j=1}^{n_2} (d_{2,j} - \bar{d}_2)^2}{n_1 + n_2 - 2}}, \quad L = \frac{\bar{d}_1 - \bar{d}_2}{s_L \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

L is approximately distributed according to a $t_{n_1+n_2-2}$ distribution, or (equivalently) L^2 is approximately distributed according to an F_{1,n_1+n_2-2} distribution.

Breusch-Pagan Test

- The Breusch-Pagan test (also known as the Cook-Weisberg score test) is an alternative to the modified Levene test. Whereas the modified Levene test is a nonparametric test, the Breusch-Pagan test assumes that the error terms are normally distributed, with $E(\epsilon_i) = 0$ and $\text{Var}(\epsilon_i) = \sigma_i^2$ (i.e., nonconstant variance). The σ_i^2 values depend on the horizontal axis quantity (X_i) values in the following way:

$$\log \sigma_i^2 = \gamma_0 + \gamma_1 X_i$$

We are interested in testing the null hypothesis of constant variance versus the alternative hypothesis of nonconstant variance. Specifically, the hypothesis test is formulated as:

$$H_0: \gamma_1 = 0$$

$$H_1: \gamma_1 \neq 0.$$

This test is carried out by first regressing the squared residuals on the predictor (i.e.,

regressing e_i^2 on X_i). The sum of squares resulting from this analysis is denoted by SSR^* , which provides a measure of the dependency of the error term on the predictor. The test statistic is given by

$$X^{2*} = \frac{SSR^*/2}{(SSE/n)^2},$$

where SSE is from the regression analysis of the response on the predictor. The p -value for this test is found using a χ^2 distribution with 1 degree of freedom (written as χ_1^2).

Bartlett's Test

Bartlett's test is another test that can be used to test for constant variance. The objective is to again perform the following hypothesis test:

H_0 : the variance is constant

H_A : the variance is not constant.

Bartlett's test is highly sensitive to the normality assumption, so if the residuals do not appear normal (even after transformations), then this test should not be used. Instead, the Levene test is the alternative to Bartlett's test that is less sensitive to departures from normality.

This test is carried out similarly to the Levene test. Once you have partitioned the residuals into g groups, the following test statistic can be constructed:

$$B = \frac{(n-g) \ln s_p^2 - \sum_{i=1}^g (n_i-1) \ln s_i^2}{1 + \left[\frac{1}{3(g-1)} \left(\left(\sum_{i=1}^g \frac{1}{n_i-1} \right) - \frac{1}{n-g} \right) \right]}.$$

The test statistic B is distributed according to a χ_{g-1}^2 distribution, so if $B \geq \chi_{g-1;1-\alpha}^2$, then reject the null hypothesis and conclude that there is statistically significant evidence that the variance is not constant.

Checking independence

A common way to test the independence assumption is to test for autocorrelation (or serial correlation) using such tests as runs test, the Durbin and Watson test.

Checking influential observations, **Checking multicollinearity (your task)**

Questions

1. What is predictive analytics?
2. What are the main types of data mining tasks?
3. What is the difference between predictive and descriptive data mining?
4. What are the six steps in predictive analytics?
5. Why is data understanding important in predictive modeling?
6. What is the difference between classification and regression in predictive analytics?
7. What challenges are commonly faced in predictive analytics?
8. What is the purpose of feature selection in predictive modeling?
9. A telecom company wants to predict which customers are likely to leave (churn) based on historical data, including customer complaints, billing, and usage patterns.
 - a. What type of predictive analytics technique would be most useful?
 - b. What data would be important for this model?
 - c. What steps should be taken in data preparation?
 - d. How would the model be evaluated?
 - e. How would the company use the model in real-time?
10. A retail company wants to predict future sales across its stores using past transaction data.
 - a. What predictive analytics technique should be used?
 - b. What key features should be considered in the model?
 - c. How can the model be validated?
 - d. How can the model be used in real-time?
11. A bank wants to predict loan defaults using customer data, including income, credit history, and spending behavior.
 - a. What type of predictive model should be used?
 - b. What features would be important?
 - c. What challenges might arise in modeling?
 - d. How can the model be deployed?
12. An e-commerce company wants to segment its customers for better-targeted marketing campaigns.
 - a. What technique is most suitable for customer segmentation?
 - b. What data should be used?
 - c. How would the company use the segmentation results?

13. A hospital wants to predict which patients are likely to be readmitted within 30 days of discharge.
- What type of predictive analytics model should be used?
 - What data would be important for the model?
 - How could predictive analytics improve hospital efficiency?
14. For the following data, your business objective, to determine the effect that price and in-store promotional expenses will have on sales, calls for examining a multiple regression model in which the price and the monthly budget for in-store promotional expenses in dollars are the independent variables and the number of OmniPower bars sold in a month is the dependent variable.

Store	Sales	Price	Promotion	Store	Sales	Price	Promotion
1	4,141	59	200	18	2,730	79	400
2	3,842	59	200	19	2,618	79	400
3	3,056	59	200	20	4,421	79	400
4	3,519	59	200	21	4,113	79	600
5	4,226	59	400	22	3,746	79	600
6	4,630	59	400	23	3,532	79	600
7	3,507	59	400	24	3,825	79	600
8	3,754	59	400	25	1,096	99	200
9	5,000	59	600	26	761	99	200
10	5,120	59	600	27	2,088	99	200
11	4,011	59	600	28	820	99	200
12	5,015	59	600	29	2,114	99	400
13	1,916	79	200	30	1,882	99	400
14	675	79	200	31	2,159	99	400
15	3,636	79	200	32	1,602	99	400
16	3,224	79	200	33	3,354	99	600
17	2,295	79	400	34	2,927	99	600

- determine whether price significantly improves the model after promotional has been included
- determine whether promotional significantly improves the model after price has been included

15. A real estate builder wishes to determine how house size (House) is influenced by family income (Income) and family size (Size). House size is measured in hundreds of square feet and income is measured in thousands of dollars. The builder randomly selected 50 families and ran the multiple regression. Partial Microsoft Excel output is provided below:

Regression Statistics	
Multiple R	0.8479
R Square	0.7189
Adjusted R Square	0.7069
Standard Error	17.5571
Observations	50

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression		37043.3236	18521.6618		0.0000
Residual		14487.7627	308.2503		
Total	49	51531.0863			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	-5.5146	7.2273	-0.7630	0.4493
Income	0.4262	0.0392	10.8668	0.0000
Size	5.5437	1.6949	3.2708	0.0020

Also $SSR(X_1 | X_2) = 36400.6326$ and $SSR(X_2 | X_1) = 3297.7917$

1. what fraction of the variability in house size is explained by income and size of family?
2. which of the independent variables in the model are significant at the 5% level?
3. when the builder used a simple linear regression model with house size (House) as the dependent variable and family size (Size) as the independent variable, he obtained an r^2 value of 1.25%. What additional percentage of the total variation in house size has been explained by including income in the multiple regression?
4. which of the following values for the level of significance is the smallest for which each explanatory variable is significant individually?

5. which of the following values for the level of significance is the smallest for which at least one explanatory variable is significant individually?
6. the value of the partial F test statistic is _____ and df _____ for
 H_0 : Variable X_1 does not significantly improve the model after variable X_2 has been included
 H_1 : Variable X_1 significantly improves the model after variable X_2 has been included
7. the value of the partial F test statistic is _____ and df _____ for
 H_0 : Variable X_2 does not significantly improve the model after variable X_1 has been included
 H_1 : Variable X_2 significantly improves the model after variable X_1 has been included
16. Test at the $\alpha = 0.05$ level to determine whether the price variable significantly improves the model given that advertising is included

ANOVA for advertising only		
	df	SS
Regression	1	17484.22249
Residual	13	39009.11085
Total	14	56493.33333

ANOVA for advertising and price			
	df	SS	MS
Regression	2	29460.02687	14730.01343
Residual	12	27033.30647	2252.775539
Total	14	56493.33333	

1. What is the Lack of Fit Test used for, and what are its assumptions?
2. What are the key formal tests for checking normality of residuals?
3. Suppose you fit a linear regression model to predict housing prices and obtain residuals. Which visualizations would you use to check assumptions of normality and homoscedasticity?

4. A regression analysis produces residuals. To test for constant variance, the Breusch-Pagan test is conducted, and a p-value of **0.02** is obtained. What do you conclude?
5. The residuals from a regression model are tested using the Shapiro-Wilk test, and the test statistic W is found to be **0.85** with a p-value of **0.03**. What does this indicate?
6. What are the hypotheses tested in the Lack of Fit test?
7. What tests can be used to check normality of residuals?
8. How is the Breusch-Pagan test used in regression analysis?
9. Why is checking residual plots important in regression analysis?
17. Given the following ANOVA table, determine if the model has a significant lack of fit.

Source	SS	df	MS	F	p-value
Regression	5,141.3	1	5,141.3	3.14	0.11
Error	14,741.6	9	1,638		
Lack of Fit	13,593.6	4	3,398	14.8	0.006
Pure Error	1,148.0	5	230		
Total	19,882.9	10			